

Lesson Exemplar in General Science

Quarter 3

Lesson

11

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Development Team

Writer: Crisanta M. Perez
Region CARAGA
Division of Dinagat Islands

Validator: Ricky F. Bosque

Language Editor:

Consultant: Crist John Pastor, PhD
Philippine Normal University

Learning Area Specialist: John Cyrus L. Doblada

Bureau of Learning Delivery
Bureau of Curriculum Development
Bureau of Learning Resources

Every care has been taken to ensure the accuracy of the information provided in this material. For inquiries or feedback, please write or call the Office of the Director of the Bureau of Learning Resources via telephone numbers (02) 8634-1072 and 8631-6922 or by email at blr.od@deped.gov.ph.



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LESSON EXEMPLAR

Learning Area	General Science	Grade Level	11
Semester	1	Quarter	3

I. OBJECTIVES (*Identifying the Goals*)

Content Standard	The learners learn that climate change affects ecosystems and the organisms living in them; and ecosystems are finite and have limited resources.
Performance Standard	By the end of the quarter, learners use examples to describe the relationships among living systems and their components and how the structure of a part relates to its function. They explain the role of cells in perpetuating life and how organ systems work together to maintain optimal health. They conduct investigations on the effects of climate change on local ecosystems.
Learning Competencies	The learners explain propose ways to adapt and mitigate the impacts of climate change on local biodiversity;
II. REFERENCES and MATERIALS (<i>Selecting Resources and Material</i>)	Alberto, Roann P., Annie Melinda Paz-Alberto, Carl Dionelle B. Ponce, and Kimberly Joy E. Mata. 2022. "Climate Change Community-Based and Ecosystem-Based Adaptation Strategies in Selected Coastal Barangays in Masinloc, Zambales, Philippines." American Journal of Climate Change 11 (04): 342–62. https://doi.org/10.4236/ajcc.2022.114017 . "Mitigating the Impacts of Climate Change on Biodiversity ZSL." n.d. Www.zsl.org. https://www.zsl.org/what-we-do/science-research/mitigating-impacts-of-climate-on-change-biodiversity .



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	“Strategies - FundsforNGOs - Grants and Resources for Sustainability.” 2024. FundsforNGOs - Grants and Resources for Sustainability - Grants and Resources for Sustainability. March 5, 2024. https://www.fundsforngos.org/proposals/strategies. 2025. Youtu.be. 2025. https://youtu.be/c8noLpZBQUs.	
III. CONTENT <i>(Sequencing Content)</i>	Adapting and Mitigating Climate Change Impacts on Local Biodiversity	
IV. OBJECTIVES <i>(Setting Clear Objectives and Analyzing the Tasks)</i>	At the end of the lesson, the learners should be able to: 1. Analyze local adaptation and mitigation practices addressing climate change impact on biodiversity; 2. Evaluate the effectiveness of these practices in conserving biodiversity. 3. Identify examples of local adaptation and mitigation practices that support biodiversity conservation. 4. Propose practical and community-based strategies to adapt to and mitigate climate change impacts on biodiversity. 5. Value the importance of individual and collective actions in protecting biodiversity.	
IV. PROCEDURES		ANNOTATION
A. Activating Prior Knowledge	The teacher must have preliminary activities every day, such as prayer, attendance checking, unfreezing, and other classroom management activities. 1. Activating Prior Knowledge	



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OPTION 1: PHOTO COLLAGE ANALYSIS

To begin the activity, the teacher prepares a set of photos showing various Philippine ecosystems such as forests, coral reefs, mangroves, and farmlands, along with images depicting local adaptation and mitigation practices like tree planting, mangrove restoration, sustainable farming, and coastal cleanups. Learners are grouped and tasked to analyze and match each ecosystem photo with the corresponding adaptation or mitigation practice. After matching, each group discusses and writes short notes explaining how the identified practice helps reduce or adapt to the impacts of climate change on that ecosystem. The groups then display their photo collages and explanations on the board or wall. The teacher concludes the activity by facilitating a discussion that highlights how these local practices support biodiversity conservation and strengthen community resilience against climate change.

OPTION 2: COMMUNITY SHARING

The teacher labels the four corners of the classroom with “Agree,” “Disagree,” “Unsure,” and “No Idea.” Learners are told that statements about biodiversity will be read aloud, and they must move to the corner that best represents their opinion. For each statement, a few learners explain their choices. The teacher listens, highlights sound reasoning, and clarifies misconceptions. This

Sample Key Question for Option 1 (after all the presentations): “Based on the photos and your group’s analysis, which ecosystem in your community do you think is most vulnerable to the impacts of climate change, and what local adaptation or mitigation practices have you observed that help protect it?”

Sample Key Question for Option 2 :

“What made you choose this corner, and how does your choice connect to what you’ve seen or experienced in your community?”



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	allows learners to reflect on their prior knowledge while actively engaging in movement and discussion. 10 Questions (Statements to Read Aloud) 1. Mangroves protect communities from strong waves and typhoons. 2. Forests only provide wood and timber products. 3. Biodiversity means having many kinds of plants and animals. 4. Coral reefs are called the “rainforests of the sea.” 5. Cutting down a few trees in a forest has no impact on biodiversity. 6. Rice fields and farms are also ecosystems. 7. Climate change has no effect on coral reefs. 8. Protecting biodiversity is important for human survival. 9. Once a species goes extinct, it can come back naturally. 10. We can help protect biodiversity even in small ways, like planting trees or reducing plastic use.	<i>Possible Answers and Clarifications</i> 1. True – Mangroves serve as natural barriers against storm surges and coastal erosion. 2. False – Forests provide oxygen, water regulation, biodiversity, and food, not just timber. 3. True – Biodiversity is the variety of life in an area. 4. True – Coral reefs are highly diverse, earning the nickname “rainforests of the sea.” 5. False – Cutting even a few trees can disturb habitats and fragment ecosystems. 6. True – Farms are ecosystems too, though man-made. 7. False – Climate change causes coral bleaching and ocean warming. 8. True – Humans depend on biodiversity for food, medicine, and resources. 9. False – Extinction is permanent; species cannot return naturally. 10. True – Small individual actions, like planting trees or reducing plastic, help biodiversity conservation.
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OPTION 3: “WHO AM I?” BIODIVERSITY RIDDLE

The teacher engages the class with a guessing game by reading short riddles or clues that describe different ecosystems. Learners listen carefully and then raise their hands or call out their guesses about which ecosystem is being described. Once the correct answer is identified, the teacher explains briefly how the clues connect to the ecosystem’s characteristics. This playful activity helps learners

Sample Key Questions for Option 3 (after the presentations): “What clue helped you identify the ecosystem, and what does it tell us about the role of that ecosystem in protecting biodiversity?”

Answer to the Riddles

1. Forest
2. Coral Reef
3. Mangrove Forest
4. Farmland/Rice Field
5. Grassland/Dry Ecosystem

Sample Key Questions for Option 4 (after the presentations): “Looking at the areas you marked, how do these environmental changes affect the people, livelihoods, or biodiversity in that place?”



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recall key concepts while creating curiosity for further discussion on biodiversity.

Five Riddles

1. I have tall trees, give you fresh air, and shelter many animals. Who am I?
2. I live under the sea, colorful and full of fish, but I bleach when waters get too warm. Who am I?
3. I grow along the shore, my roots hold the soil, and I protect people from strong waves. Who am I?
4. I cover wide land, planted by farmers, and I provide food for people every day. Who am I?
5. I have little rain, my soil is dry, and only a few plants and animals can survive in me. Who am I?

OPTION 4: LOCAL MAP MARKING

The teacher presents a large map of the province or community and asks learners to think about areas where they have observed environmental or climate impacts, such as flooding, drought, or deforestation. Working in groups, learners mark or place symbols on the map to represent these observations. Afterward, each group

Sample Key Questions for Option 5 (after the presentations): “How does this problem affect both the ecosystem and the people who depend on it?”



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briefly shares their markings and explanations. The teacher affirms correct inputs, supplements with additional examples, and connects the activity to the broader topic of ecosystems and climate change.

OPTION 5: ECOSYSTEM CHARADES

The teacher prepares slips of paper with different ecosystem problems such as deforestation, coral bleaching, or drought. Learners take turns drawing a slip and acting out the problem silently while classmates guess the scenario. Once the correct answer is given, the teacher confirms and briefly explains the issue. This active and fun activity encourages learners to express their understanding through movement and reinforces awareness of real environmental challenges.

2. Establishing the Purpose of the Lesson

The teacher begins by asking learners:

- “What changes in nature have you noticed in your community over the past years—like in the weather, the sea, or the forests?”



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	• “How do these changes affect the plants, animals, or even the way people live in your area?” • Learners are then asked to imagine their community in the future and reflect on the question: • “If these changes continue, what challenges might your family, neighbors, or future work face?” The teacher then explains that today’s lesson will help them understand how climate change affects biodiversity, explore ways communities can adapt and mitigate its impacts, and think of practical strategies they can do individually and together to protect ecosystems. Finally, the teacher presents the lesson objectives for the learners to read and reflect on.	
B. Instituting New Knowledge	1. Presenting Examples To connect the lesson to what learners already know, the teacher reminds them that in earlier grades they studied how living things depend on each other through food chains, habitats, and simple ecosystem relationships. This time, they will go further by examining how these ecosystems are being affected by climate change and why these matters for both biodiversity and human communities. The teacher presents several examples of Philippine ecosystems—forests, coral reefs, mangroves, farmlands, and rivers—using photos or short descriptions to highlight their unique roles. Learners are encouraged to look closely at these examples and identify the benefits people gain from them, such as clean water, food, protection	<i>If the learners are unable to provide the appropriate definitions of the difficult ecological terms (e.g., biodiversity, coral bleaching, habitat degradation), the teacher should step in to give clear and accurate explanations. Suggested definitions may include:</i> Biodiversity refers to the variety of life in a particular habitat or ecosystem. Habitat degradation refers to the process by which a natural habitat becomes incapable of supporting its species.



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from floods, or sources of livelihood. The teacher then points out that when ecosystems are damaged by climate change—like coral bleaching, stronger typhoons, or prolonged drought—these benefits are also threatened. Learners are asked whether they have seen changes in these ecosystems in their own area or heard stories from their community about such changes. The teacher clarifies new or difficult terms that appear in the examples and encourages peers to explain in their own words to promote understanding. Finally, the teacher emphasizes that this lesson will help them not only describe these impacts but also think of adaptation and mitigation strategies, and recognize the value of both individual and collective actions in protecting biodiversity. ACTIVITY 1: VIDEO CLIP + GROUP REACTION Procedure: 1. Teacher shows a short video clip about a local biodiversity-related action (e.g., mangrove planting, reef restoration, reforestation). 2. Learners watch attentively, taking notes on what they observe. 3. In small groups, learners discuss their reactions to the video (What happened? Why was it done? What results were seen?). 4. Groups share their key points in class.	Coral bleaching refers to the whitening of corals due to stress factors such as warmer ocean temperatures. <i>The teacher ensures that learners have a basic understanding of these concepts before moving forward. To check comprehension, learners may be asked to give examples of these terms in the context of Philippine ecosystems. The teacher may modify the examples or add more depending on the learners' level of familiarity.</i> <i>They can access this link:</i> https://youtu.be/hmDUvwUN9HI <i>The video helps learners see real examples of how ecosystems and communities are connected, and how local actions like mangrove planting can both protect biodiversity and address climate change. Their reactions give the teacher a chance to clarify misconceptions and connect concrete</i>
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	5. The teacher let the learners connect the discussion to how such actions contribute to adaptation or mitigation. Processing Questions: - What problems did the video show, and how were they addressed? - How does planting mangroves help both people and ecosystems? - Do you consider this an adaptation strategy, a mitigation strategy, or both? Why? - How can similar practices be applied in your own community? ACTIVITY 2. TIMELINE DRAWING Procedure: - The teacher distributes paper and markers. - Learners are asked to imagine one local ecosystem (e.g., coral reef, forest, river, farmland). - Groups draw a timeline showing how the ecosystem might change over 20 years if climate stress (heat, storms, drought, floods) continues.	<i>observations to abstract concepts like adaptation and mitigation.</i> <i>Possible answer to Processing Questions:</i> <i>The problem was flooding or erosion, and planting mangroves helped reduce the damage.</i> <i>Mangroves provide breeding grounds for fish, protect coasts from strong waves, and prevent soil erosion. They also trap carbon, which helps lessen climate change.</i> <i>Some may say mangrove planting is adaptation because it protects people, others may say it is mitigation because it reduces greenhouse gases, and some may argue it is both.</i> <i>Similar practices in the community could include tree planting along rivers, protecting watershed areas, urban gardening, or joining coastal clean-ups.</i> <i>This activity helps learners visualize how ecosystems gradually change under climate stress. By drawing timelines, they see the</i>
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4. Groups present their drawings to the class, highlighting key changes.
5. The teacher guides the class in comparing patterns and emphasizes long-term impacts of climate change.

Processing Questions:

- a. What changes did you draw for the first 5 years? For the next 10? After 20 years?
- b. How do these changes affect people who depend on this ecosystem?
- c. Which changes are reversible, and which may be permanent?
- d. What actions could prevent or lessen these changes?

ACTIVITY 3. GROUP SORTING (ADAPTATION VS. MITIGATION)

Procedure:

1. The teacher prepares index cards with various actions written on them (e.g., planting trees, building seawalls, using renewable energy, elevating houses, reducing plastic, practicing organic farming).
2. Groups are asked to sort each card under Adaptation (adjusting to climate impacts) or Mitigation (reducing the causes of climate change).
3. Each group explains how they sorted the cards.
4. The teacher confirms correct placements and clarifies confusing cases.

cumulative impact of climate change and realize which changes are reversible and which are permanent.

Possible answer to Processing Questions:

- a. *Forests may lose trees within 5 years, wildlife declines after 10 years, and soil becomes barren after 20 years. Coral reefs may fade and bleach within 5 years, fish populations decline by 10 years, and dead reefs dominate after 20 years.*
- b. *Rivers may dry up due to drought, affecting farms and water supply.*
- c. *Changes could make farming or fishing harder and reduce food supply.*
- d. *Preventive actions include tree planting, sustainable farming, reducing pollution, and protecting reefs.*

Sorting activities help learners differentiate between adaptation and mitigation in practical ways. It allows them to test their understanding, defend their reasoning, and see overlaps between strategies.



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	Processing Questions: - Why did you place this action under adaptation? Under mitigation? - Are their actions that can be considered for both? Why? - Which of these actions can be done by individuals? Which require community or government support? - Which strategies do you think are most urgent in your area? ACTIVITY 4. “WHAT IF?” SCENARIOS Procedure: - Teacher presents hypothetical scenarios such as: “What if forests in Mindanao are lost to fire?” “What if coral reefs in Palawan disappear due to bleaching?” “What if droughts damage rice fields in Central Luzon?” - Learners work in groups to predict the ecological, social, and economic impacts of each scenario. - Groups share their answers. - The teacher synthesizes and links to biodiversity loss and	<i>Possible answers to Processing Questions:</i> <i>Planting trees can be both adaptation (providing shade, preventing floods) and mitigation (absorbing carbon).</i> <i>Building seawalls and elevating houses are adaptation actions. Using renewable energy and waste segregation are mitigation actions.</i> <i>Some actions can be done by individuals (e.g., home gardening, reducing plastic use), while others need community or government support (e.g., renewable energy, reforestation).</i> <i>Learners may identify urgent actions such as coastal protection, water conservation, and forest protection.</i> <i>By predicting the consequences of climate-related disasters, learners practice systems thinking—seeing how the loss of one ecosystem affects people, biodiversity, and the economy.</i>
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	climate change impacts. Processing Questions: - What will happen to biodiversity in the given scenario? - How will communities and livelihoods be affected? - What chain reactions might follow in other ecosystems? - What adaptation or mitigation strategies could reduce these risks? ACTIVITY 5. READING CIRCLES Procedure: - Teacher prepares short case briefs (e.g., community tree planting, reef rehabilitation, renewable energy project, water conservation initiative). - Class is divided into small groups; each group receives one case brief. - Within groups, learners take turns reading parts of the case. - After reading, they discuss the main points and prepare a short summary. - Each group presents their summary to the class. - The teacher highlights key lessons from each case.	<i>Possible answers to Processing Questions:</i> <i>If forests in Mindanao are lost to fire, species will go extinct, soil erosion will worsen, and flooding risks increase. If coral reefs in Palawan disappear, fish populations collapse, tourism declines, and fisherfolk lose livelihoods.</i> <i>If drought damages rice fields, food supply drops, prices rise, and farming families suffer.</i> <i>Ecosystem loss leads to chain reactions such as poverty, migration, and loss of cultural practices.</i> <i>Adaptation strategies may include drought-resistant crops or community disaster planning; mitigation includes reducing greenhouse gases and protecting natural resources.</i> <i>Reading case studies allows learners to analyze real community practices. Working in groups strengthens collaboration and ensures that complex information is simplified through peer discussion.</i>
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	Processing Questions: - What was the main problem described in the case? - What adaptation or mitigation strategy was used? - What were the positive effects on biodiversity and people? - Could this strategy be applied in your community? Why or why not? Checkpoint: The teacher asks the learners what new terms and concepts they have learned so far based on the activities. 2. Discussing New Concept - The teacher begins with a storytelling prompt: “Imagine our community 20 years from now. The sea is warmer, storms are stronger, and some plants and animals are disappearing. What changes might you see around you?” Learners briefly share their thoughts, connecting their imagination to real-world issues.	<i>Possible answers to Processing Questions:</i> <i>A case on tree planting may highlight reduced flooding and increased biodiversity.</i> <i>A case on reef rehabilitation may show fish returning and better catches for fisherfolk. A renewable energy project may reduce carbon emissions and lower household costs.</i> <i>A water conservation initiative may ensure safe drinking water during dry seasons.</i> <i>Learners may say these strategies are useful in their own community, such as planting trees near rivers, starting waste segregation, or practicing sustainable fishing.</i>
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2. The teacher introduces the idea that climate change affects ecosystems differently, depending on their location and features. Instead of posting a chart immediately, the teacher presents photos or short case examples of local ecosystems (e.g., a bleached coral reef, a dried farmland, a flooded mangrove area, a deforested hillside). Learners are asked: “What do you notice? What do you think caused this change?”

3. Learners are grouped, and each group is given one ecosystem case (e.g., forest, reef, mangrove, freshwater, farmland, urban). They are tasked to discuss:

- How does climate change affect this ecosystem?
- How might the loss of this ecosystem affect people’s lives?

4. Groups then share their answers. As each group presents, the teacher highlights scientific explanations (e.g., coral bleaching due to rising sea temperatures, soil erosion due to stronger typhoons, urban heat islands due to loss of trees).

5. The teacher writes emerging key terms on the board (e.g., coral bleaching, biodiversity loss, drought, flooding, habitat degradation).

6. Learners are invited to try explaining these terms in their own words before the teacher provides the accurate definition.

Finally, the teacher summarizes by connecting back to the lesson objectives: ecosystems are interconnected, climate change threatens biodiversity, and communities must respond through adaptation and mitigation.

The teacher uses this checkpoint to monitor learners’ grasp of new vocabulary and concepts (e.g., ecosystem services, coral bleaching, biodiversity loss).



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Activity 6: TWO-COLUMN NOTES

The teacher begins by asking learners to think about how people respond to climate change: “Do we adjust to the changes, or do we try to reduce the causes of the changes?” The teacher then introduces the two approaches—**adaptation** and **mitigation**—and explains that both are important for protecting biodiversity. On the board, the teacher draws a large chart with two columns labeled **Adaptation** and **Mitigation**. Learners copy the same chart in their notebooks. To model the task, the teacher gives one example for each side (e.g., building seawalls under Adaptation and planting trees under Mitigation).

Next, the teacher presents a set of local action examples—such as using solar panels, planting mangroves, shifting to drought-resistant rice, practicing waste segregation, building evacuation centers, and conserving energy at home. Learners are asked to decide whether each action is an adaptation or a mitigation strategy. As each example is placed in the class chart, selected learners explain their reasoning. The teacher guides the discussion, clarifies misconceptions, and highlights that some actions can serve both adaptation and mitigation. Finally, learners are tasked to add **one example from their own community** under each column, which is then shared with the class. The activity concludes with the teacher summarizing that adaptation helps us adjust to today’s impacts, while mitigation prevents worse impacts in the future—essential in conserving biodiversity.

Clarifications the Teacher May Add

On Overlaps: Some actions (like mangrove planting or tree planting) are both adaptation and mitigation. Clarify that strategies can serve multiple purposes depending on their impact.

On Timescale:

Adaptation is often short- to medium-term (e.g., building shelters, adjusting crops).

Mitigation is often long-term (e.g., reducing carbon emissions, restoring forests).

On Focus:

Adaptation = protects people and ecosystems from current/future risks.

Mitigation = addresses the root cause by reducing greenhouse gases.



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3. Developing Mastery

ACTIVITY 7 (OPTION 1): FISHBOWL DISCUSSION

The teacher arranges the class into two circles: an inner circle (5–7 students) and an outer circle (the rest). The inner circle is tasked to discuss the guiding question: *“Is adaptation more urgent than mitigation in addressing climate change impacts on biodiversity?”* They share opinions, cite examples, and respond to each other, while the outer circle listens and takes notes. After a set time, roles are switched so more students can participate. At the end, the teacher invites volunteers from the outer circle to summarize what they heard before offering clarifications.

Processing Prompts & Possible Answers

- Some learners may argue adaptation is more urgent because people need protection now (e.g., seawalls, evacuation centers, drought-resistant crops).
- Others may say mitigation is equally or more urgent to stop climate change from worsening (e.g., renewable energy, reforestation).
- Learners might conclude that both are necessary, but communities facing immediate risks may prioritize adaptation.

On Local Connection: Give real Philippine examples (e.g., seawalls in Leyte after Typhoon Yolanda, solar panel projects in Palawan, mangrove planting in Cavite). This strengthens relevance and ensures learners see that adaptation and mitigation are not abstract—they are happening in their country.

This activity develops critical thinking and debate skills while deepening understanding of adaptation vs. mitigation. It highlights that the two are not opposing, but complementary. Clarify that urgent adaptation saves lives today, while long-term mitigation prevents future crises. Encourage respect for differing opinions while grounding arguments in evidence and local realities.



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ACTIVITY 7 (OPTION 2): TRUE/FALSE RELAY

The teacher divides the class into groups. Each group lines up at the back of the classroom. The teacher shows a flashcard with a statement about ecosystems, biodiversity, or climate strategies. One student from each team runs to a designated area and writes True or False on the board (or raises a sign). Correct answers earn points for the team. The game continues until all statements are answered.

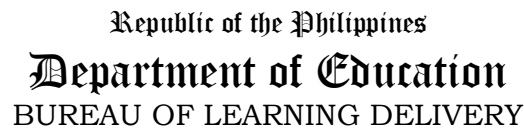
True/False Relay Statements

1. Coral reefs bleach when sea temperature rises.
2. Planting mangroves is only a mitigation strategy.
3. Adaptation means reducing greenhouse gases.
4. Urban heat islands are caused by fewer trees and more concrete.
5. Deforestation increases the risk of flooding.
6. Mitigation strategies give results immediately.
7. Using drought-resistant rice varieties is an adaptation strategy.
8. Installing solar panels is an adaptation strategy.
9. Biodiversity loss can reduce people's sources of food and medicine.
10. Building seawalls protects communities from storm surges, so it is a mitigation strategy.

ACTIVITY 7 (OPTION 3): MINI-POSTER MAKING

Answer Key

1. *True*
2. *False – it's both adaptation and mitigation*
3. *False – that's mitigation*
4. *True*
5. *True*
6. *False – they usually take years or decades to show effects*
7. *True*
8. *False – it's mitigation because it reduces fossil fuel use*
9. *True*



General Science Quarter 3



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C. Demonstrating Knowledge and Skills	The teacher explains that understanding ecosystems and climate change is only the first step. What truly matters is how learners can use this knowledge in real situations: helping farmers adapt to drought, supporting fisherfolk when reefs are damaged, or working with neighbors to prevent flooding. Through the next activities, learners will practice applying adaptation and mitigation strategies to solve community problems, preparing them to make responsible choices in both their daily lives and future careers. 1. Finding Practical Applications ACTIVITY 8 OPTION 1: CASE STUDY PROBLEM-SOLVING 1. The teacher presents short case studies about different communities facing climate change impacts (e.g., a coastal barangay losing mangroves, a farming village experiencing drought, or an urban community facing heat waves and flooding). Each group is assigned one case and asked to identify the main problem, then propose both an adaptation strategy and a mitigation strategy for the situation. Groups record their solutions on chart paper and present them to the class. The teacher highlights similarities and differences across solutions and clarifies misconceptions. Processing Prompts & Possible Answers What is the community's main problem? → Flooding, drought, heat stress, loss of fish catch, deforestation.	<i>This activity develops problem-solving skills and applies concepts directly to real-world situations. Clarify that adaptation answers “How do we cope?” while mitigation answers “How do we prevent worse impacts?”. Stress that communities often need to apply both at once.</i>
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	What adaptation strategy can help? → Building seawalls, shifting to drought-resistant crops, community rainwater harvesting, coastal evacuation plans. What mitigation strategy can help? → Planting mangroves, tree planting, promoting renewable energy, reducing waste. Who benefits from these solutions? → Farmers, fisherfolk, students, families, local businesses. ACTIVITY 8 OPTION 2: ROLE-PLAY CASE RESPONSES 1. Each group is assigned a stakeholder role (farmers, fisherfolk, youth, barangay officials). The teacher presents a case (e.g., stronger typhoons damaging farms). Groups discuss and act out their responses, showing what their stakeholder would propose as solutions. Processing Questions a. What solutions would your stakeholder prioritize? b. How are the needs of different stakeholders connected? ACTIVITY 8 OPTION 3: IF-THEN” CAUSE-EFFECT CHAINS	<i>T</i>
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	Procedure Groups are given a situation (e.g., mangrove loss, deforestation, heat waves). They create a flowchart: If climate change causes __, then __ happens to the ecosystem, leading to __ for the community. The adaptation is __, and mitigation is __. Groups present their chain on the board. Processing Questions: - What effect does climate change have on this ecosystem? - How does it affect people's lives? - What are both adaptation and mitigation responses? Learners share their reflections in pairs, then volunteers share with the whole class. The teacher emphasizes that understanding ecosystems is not only academic—it's connected to real future opportunities. ACTIVITY 9 OPTION 1: "BARANGAY COUNCIL SIMULATION" The Learners role-play as barangay councilors tasked with creating 3 eco-strategies. Groups draft policies, ensuring at least one adaptation and one mitigation measure. They present to the class as if in a barangay assembly.	<i>Possible</i> <i>Answers:</i> <i>If mangroves are lost → then coasts are exposed → leading to flooding → adaptation: build seawalls, plant mangroves → mitigation: reduce cutting of trees.</i>
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	Processing Questions: - What problems in your barangay are most urgent? - Which strategies will help adapt? Which will mitigate? - Why should the community support your proposal? ACTIVITY 9 OPTION 2: YOUTH POLICY PROPOSALS	<i>Possible Learner Responses</i> <i>Eco-strategies: Waste segregation, mangrove planting, solar streetlights.</i> <i>Adaptation: Flood warning system.</i> <i>Mitigation: Tree planting to absorb carbon.</i> <i>Possible Responses:</i> <i>Youth-led clean-up drives.</i>
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	The Groups act as youth representatives tasked with submitting proposals at a barangay meeting. They brainstorm 2–3 eco-strategies that focus on youth participation. They present their proposals in front of peers. Processing Questions: - How can young people contribute to adaptation? - How can youth lead mitigation efforts? - Why is youth involvement important? The teacher summarizes by emphasizing interconnectedness: climate change → ecosystems → people. ACTIVITY 10: SIMPLE 3-STEP TEMPLATE The learners individually or in pairs fill out the chart: Problem → Solution → Who should act. After filling out, they share with the class. Processing Questions - What local problem is urgent? - What is your suggested solution?	<i>School waste segregation campaigns.</i> <i>Mangrove planting as community service.</i> <i>Possible Learners Responses:</i> <i>Problem: Flooding in barangay road. → Solution: Clean drainage, plant trees. → Who: Barangay council + residents.</i>
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BUREAU OF LEARNING DELIVERY

c. Who should lead the action?

ACTIVITY 11 OPTION 1: JOURNALING

Learners write a short journal entry reflecting on how climate change might affect their family's livelihood, job, or community role (e.g., farming, fishing, tourism, small business). They also write what adaptation or mitigation action could help. Volunteers may share excerpts with the class.

Processing Prompts

- How might climate change affect your family's work or income?
- What strategies could protect or improve your livelihood?

ACTIVITY 11 OPTION 2: CLIMATE CHANGE TREE

The teacher draws a large tree outline on the board or Manila paper. Learners are given colored sticky notes or cut-outs. They write:

On the trunk: Climate change problems affecting biodiversity.

On the branches: Possible solutions (adaptation/mitigation).

Learners can access the following links to read:

Possible Outcomes (answers):

"If drought continues, our rice fields will suffer, but planting drought-resistant crops can help."

"Our family depends on fishing, but coral bleaching reduces fish. Marine sanctuaries could help restore them."

"Tourism in our place might decline if forests are destroyed, but reforestation could bring it back."



Republic of the Philippines
Department of Education
BUREAU OF LEARNING DELIVERY

	On the leaves: Personal actions they can commit to. Students post their notes on the tree, creating a collaborative “class climate action tree.” Processing Prompts - What are the biggest problems we face due to climate change? - What solutions can communities or leaders apply? - What personal actions will you commit to? 2. Making Generalization ACTIVITY 12 OPTION 1: CONCEPT WEB The teacher guides the class in creating a concept web with “Climate Change + Biodiversity” at the center. Learners brainstorm related ideas they learned in the lesson, such as ecosystems, impacts, adaptation, mitigation, and human survival. As they share, the teacher organizes the words in circles around the central idea and draws arrows to show relationships. For example: “Climate change leads to coral bleaching” (impact), “Mangrove planting is both adaptation and mitigation,” or “Protecting ecosystems ensures food security.” After the web is completed, the teacher leads the class in formulating a general statement that sums up the lesson: that biodiversity is directly affected by climate change, but humans can respond through adaptation and mitigation to ensure survival.	<i>Possible Outcomes:</i> <i>Trunk (Problems): Flooding, drought, deforestation, coral bleaching.</i> <i>Branches (Solutions): Reforestation, mangrove planting, waste management, renewable energy.</i> <i>Leaves (Actions): Bringing eco-bags, segregating waste, joining clean-up drives</i>
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Republic of the Philippines
Department of Education
BUREAU OF LEARNING DELIVERY

ACTIVITY 12 OPTION 2: THINK-PAIR-SHARE

The teacher asks learners to take one minute to think quietly about the question: “What is the most important connection between climate change and biodiversity?” Learners then pair up to share their answers with a partner, before a few volunteers share with the whole class. The teacher lists key words from their answers on the board (e.g., “ecosystem,” “adaptation,” “mitigation,” “livelihood”). Finally, the class works together to create one or two summary sentences that generalize the lesson’s main idea.

3. Evaluating Learning

ACTIVITY 13: “CLIMATE CHANGE AND BIODIVERSITY

CHECK”

Learners individually answer the following in their notebooks (words, phrases, or short sentences are acceptable):

1. How does climate change affect biodiversity in the Philippines?
2. What is the difference between adaptation and mitigation strategies?
3. Give one example of a local adaptation practice and one example of a local mitigation practice.

Generalization Statement (Sample):

“Climate change threatens biodiversity, but by practicing both adaptation and mitigation strategies, we can protect ecosystems and secure human survival.”

Generalization Statement (Sample):

“Climate change affects ecosystems and biodiversity, but through adaptation and mitigation, communities can protect their environment and livelihoods.”



Republic of the Philippines
Department of Education
 BUREAU OF LEARNING DELIVERY

	4. Why is it important for communities to adapt and mitigate the impacts of climate change? 5. As a student, what concrete action can you do to help protect biodiversity? Afterward, the teacher collects and checks the answers. Learners with incomplete or confused responses may be paired with classmates who answered correctly for peer clarification, before final checking. The following activity is optional and may be conducted if there is extra time beyond the teacher's allocated budget for this unit or lesson. Since it directly applies the concepts of climate change impacts on Philippine ecosystems, it may also serve as a meaningful performance task. This allows learners to extend their understanding by connecting classroom knowledge to real-world environmental issues and community actions. 4. Additional Activities	<i>Possible Responses:</i> <i>Rising sea temperatures cause coral bleaching; stronger typhoons damage forests and farms; drought reduces water and crops; sea level rise destroys mangroves.</i> <i>Adaptation = coping with effects (e.g., seawalls, drought-resistant crops); Mitigation = reducing emissions (e.g., reforestation, renewable energy).</i> <i>Adaptation: salt-tolerant rice; Mitigation: tree planting.</i> <i>Protects lives, food, water, and biodiversity; reduces disaster risks.</i> <i>Plant trees, segregate waste, save energy, reduce plastic use.</i>
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Republic of the Philippines
Department of Education
BUREAU OF LEARNING DELIVERY

ACTIVITY 14 OPTIONAL: ECO-VLOG TASK

Procedure (Steps):

1. The teacher explains the task: learners will create a 1-minute video blog (vlog) showing a local practice that protects biodiversity or helps address climate change. Examples may include planting trees, maintaining home gardens, waste segregation, mangrove planting, using eco-bags, biking instead of using motorbikes, or joining community clean-ups.
2. Learners list down possible practices they can document in their home, school, or barangay. The teacher reminds them to choose practices they can safely and realistically show.
3. Partners outline their video with three simple parts:
Introduction: State the problem or climate issue (e.g., “Plastic waste harms our rivers”).

Practice Shown: Demonstrate or show the eco-friendly action (e.g., “Here we segregate plastics and compost biodegradable waste”).

Closing Message: Explain why this practice is important for protecting biodiversity and how others can join.
4. Learners record their short video using a phone or other available device. They are encouraged to keep it clear, simple, and within 1 minute.



Republic of the Philippines
Department of Education
 BUREAU OF LEARNING DELIVERY

	5. Learners may present their vlogs in class, upload them to a shared platform (if available), or compile them into a “class eco-vlog collection.” 6. After viewing some videos, the teacher asks: “What did you learn from your classmates’ eco-vlogs that you can also apply in your own life?” Evaluation Criteria (Rubric): • Content (40%) – The vlog clearly shows a biodiversity-friendly practice and explains its importance. • Clarity (20%) – The message is easy to understand; within 1 minute. • Creativity (20%) – The vlog is engaging, original, and demonstrates effort in presentation. • Relevance (20%) – The practice shown is realistic, local, and connected to climate change and biodiversity conservation.	
V. ASSESSMENT <i>(Assessing Learnings)</i>	Two options are given in this part: a performance-based assessment and a paper-and-pen assessment. The teacher may opt to implement both at his/her discretion and based on availability of time and the needs of the learners. ASSESSMENT OPTION 1: COLLABORATIVE PERFORMANCE-BASED ASSESSMENT (“COMMUNITY CLIMATE-BIODIVERSITY ACTION CHALLENGE”) The teacher announces this activity as the summative performance task for the lesson. Learners will be assessed on both the process (collaboration, planning) and the product (clarity and feasibility of their eco-solution design).	



Republic of the Philippines
Department of Education
 BUREAU OF LEARNING DELIVERY

Climate-Biodiversity Action Challenge. The teacher explains that the purpose of this activity is for students to apply what they learned about ecosystems, climate change impacts, and adaptation and mitigation strategies by designing a practical community-based project. Learners are grouped into teams of 4–6 members and asked to select or are assigned a Philippine ecosystem such as mangroves, coral reefs, forests, rivers, farmlands, or grasslands. Each group identifies a specific climate change threat affecting their chosen ecosystem—for example, coral bleaching due to rising sea temperatures, flooding from deforestation, or drought affecting agricultural land.

The group then designs a community eco-solution that proposes concrete adaptation or mitigation strategies. The design should explain how the solution will benefit both the ecosystem and the local community, such as protecting biodiversity, supporting livelihoods, or reducing disaster risks. Their eco-solution should be presented creatively in one of the following formats: a poster, infographic, or a simple written project plan using bond paper or manila paper (digital outputs are also allowed, depending on resources).

After completing their outputs, groups exchange their work with another team for peer feedback to improve clarity, feasibility, and sustainability. Finally, each group presents their eco-solution to the class in a 3-minute presentation, followed by a **2-minute Q&A or defense session**. This process not only evaluates learners’ understanding of climate change and biodiversity but also develops teamwork, creativity, and critical thinking skills.

The teacher may use and modify the following rubric after presenting it to the class.

Criteria	Excellent (4 pts)	Proficient (3 pts)	Developing (2 pts)	Beginning (1 pt)
Content Accuracy	Shows deep understanding of ecosystem, threat, and strategies; all details accurate.	Shows clear understanding; minor inaccuracies present.	Shows partial understanding; some key points missing.	Shows little understanding; many inaccuracies.
Clarity of Eco-Solution	Eco-solution is very clear, realistic, and well explained.	Eco-solution is mostly clear; explanation is understandable.	Eco-solution is vague or lacks detail.	Eco-solution is confusing and incomplete.
Creativity & Presentation	Highly creative, engaging, and visually appealing presentation.	Presentation is neat and somewhat engaging.	Presentation shows little creativity or effort.	Minimal effort; lacks creativity or appeal.
Relevance & Feasibility	Solution is highly relevant to climate-biodiversity issues	Solution is relevant and somewhat	Solution is somewhat relevant but	Solution is not relevant or feasible.



Republic of the Philippines
Department of Education
BUREAU OF LEARNING DELIVERY

ASSESSMENT OPTION 2: MULTIPLE CHOICE TEST

Directions: Read each item carefully. Choose the letter of the correct answer.

1. A coastal town notices that fewer fish are being caught near coral reefs. Which is the most likely link to climate change?
 - a. Coral reefs are growing faster.
 - b. Typhoons increase fish populations.
 - c. Warmer seas attract more marine life.
 - d. Coral bleaching reduces reef habitats.
2. Which situation best shows an adaptation strategy?
 - a. Banning single-use plastics.
 - b. Using renewable energy in factories.
 - c. Expanding tree plantations for carbon storage.
 - d. Building houses on stilts in flood-prone areas.
3. A community plants mangroves along the coast. Why is this both adaptation and mitigation?
 - a. It improves tourism only.
 - b. It only protects fish habitats.
 - c. It cools the weather but harms biodiversity.
 - d. It reduces flooding risks and absorbs carbon dioxide.



Republic of the Philippines
Department of Education
BUREAU OF LEARNING DELIVERY

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| | <ol style="list-style-type: none">4. A farmer wants to adapt to frequent droughts caused by climate change. Which choice is best?<ol style="list-style-type: none">a. Using more chemical fertilizers.b. Planting drought-resistant crops.c. Switching to water-intensive crops.d. Expanding farmland into nearby forests.5. Which statement best explains why protecting ecosystems supports human survival?<ol style="list-style-type: none">a. Ecosystems only provide scenery for people.b. Biodiversity has no direct effect on communities.c. Climate change only affects animals, not humans.d. Healthy ecosystems provide food, water, and protection.6. Which cause-effect chain best describes the impact of deforestation in upland forests?<ol style="list-style-type: none">a. Logging → more jobs → stable climateb. More trees → healthier soil → less floodingc. Fewer trees → soil erosion → flooding in lowlandsd. Deforestation → more farmland → increased biodiversity7. A barangay council wants to design a project that reduces greenhouse gas emissions. Which plan is best?<ol style="list-style-type: none">a. Raising houses in flood zones.b. Distributing drought-resistant seeds.c. Building seawalls and evacuation centers.d. Installing solar streetlights and promoting waste segregation.8. Why might adaptation be considered urgent for vulnerable communities?<ol style="list-style-type: none">a. It is easier than mitigation.b. It stops global warming permanently.c. It reduces greenhouse gases immediately.d. It helps people cope with impacts already happening.9. During a class discussion, a student says, “Mitigation is more important than adaptation.” Which argument best challenges this?<ol style="list-style-type: none">a. Mitigation is only for rich countries. |
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Republic of the Philippines
Department of Education
BUREAU OF LEARNING DELIVERY

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| | <ul style="list-style-type: none">b. Mitigation is cheaper than adaptation.c. Adaptation reduces global emissions faster.d. Adaptation helps communities survive current disasters. <p>10. Which scenario shows a community-based adaptation practice?</p> <ul style="list-style-type: none">a. Hunting more wildlife for food security.b. Expanding plastic production factories.c. Opening new coal power plants for electricity.d. Planting trees in a watershed to reduce flooding. <p>11. A typhoon destroyed coastal mangroves. Which long-term effect may follow?</p> <ul style="list-style-type: none">a. More farmland created.b. Stronger coral reef growth.c. Improved biodiversity inland.d. Increased flood risks and loss of fish habitats. <p>12. A farmer's group proposes to use organic fertilizer and composting. Which type of strategy is this?</p> <ul style="list-style-type: none">a. Purely mitigationb. Purely adaptationc. Both adaptation and mitigationd. Neither adaptation nor mitigation <p>13. Why are urban heat islands a concern for biodiversity and people?</p> <ul style="list-style-type: none">a. They increase tree cover in cities.b. They prevent flooding during typhoons.c. They reduce electricity demand in summer.d. They trap heat, worsen air quality, and stress ecosystems. <p>14. A town faces sea level rise. Which option balances both adaptation and mitigation?</p> <ul style="list-style-type: none">a. Constructing coal plants while raising houses.b. Building seawalls while promoting solar power.c. Using diesel generators and evacuation centers.d. Expanding logging while planting water-intensive crops. |
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Republic of the Philippines
Department of Education
 BUREAU OF LEARNING DELIVERY

	15.If you were tasked to propose a community eco-solution, which plan would be most sustainable? - Build a seawall only. - Switch all farms to chemical fertilizers. - Depend on relief goods after every disaster. - Launch a reforestation project with disaster education.
VI. REFLECTION <i>(Feedback and Continuous Improvement)</i>	Learners' Reflection Ask the learners to complete the following prompts in their notebooks or on a sheet of paper. They may provide short or extended responses: The most important thing I understood about climate change and biodiversity is _____. One adaptation or mitigation strategy I believe is effective in protecting ecosystems is _____ because _____. A simple action I can take in my family or community to help address climate change is _____. I realized that if our ecosystems are not protected, _____. I still want to learn more about _____ so I can better understand how to act. Teacher's Reflection - Were the objectives achieved? What evidence shows that learners can explain climate change impacts, distinguish adaptation from mitigation, and propose strategies? - Which activities most helped learners connect concepts to real-life issues in their communities? (e.g., case study problem-solving, poster-making, policy pitch). - What difficulties did learners face? (e.g., differentiating adaptation vs. mitigation, identifying local examples, proposing realistic strategies). How did I address these?



Republic of the Philippines
Department of Education
BUREAU OF LEARNING DELIVERY

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| | 4. How can I refine the lesson to encourage deeper critical thinking, stronger local relevance, and more concrete action planning in future sessions? |
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Prepared by:

CRISANTA M. PEREZ

Special Science Teacher I
Sibagat National High School of Home Industries
Agusan del Sur Division
Caraga Region

Reviewed by:

ALFONS JAYSON O. PELGONE, PhD

Associate Dean
Philippine Normal University

Approved by:

JOHN CYRUS L. DOBLADA

Specialist, General Science
BLD-TLD

Annex A. **DESIGN THINKING PROCESS**

Project Title	
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Republic of the Philippines
Department of Education
 BUREAU OF LEARNING DELIVERY

Project Proponents	
Project Duration	

What knowledge, skills, and attitudes (KSAs) are needed in this project?

Knowledge	Skills	Attitudes

EMPATHIZE Understand the context and the need. What glaring need or problem have you observed? Why is this a problem?	
DEFINE Define the scope of your problem. What problem do you commit to address?	
IDEATE Think of potential solutions. What are your ideas on how the problem will be solved?	Will these ideas work? How?
Idea 1 	
Idea 2 	



Republic of the Philippines
Department of Education
BUREAU OF LEARNING DELIVERY

Idea 3	
PROTOTYPE/PRODUCE Create your solution. How would you translate your idea into a product, process, or output?	
TEST AND IMPLEMENT Implement your solution. How would you test if your idea is working? How would you roll out the solution?	
LEARN Evaluate your learning experience. What were your essential learning in the process of designing? Answer by highlighting the KSAs. How would this project impact you and the community if implemented? How can the project be improved?	



Republic of the Philippines
Department of Education
 BUREAU OF LEARNING DELIVERY

Annex B. **RATIONALIZATION OF THE MULTIPLE-CHOICE ASSESSMENT**

Item No.	Correct Answer	Rationalization
1	D	Coral bleaching happens when sea temperature rises, killing reef habitats that support fish. This explains why fish catch declines.
2	D	Building houses on stilts is adaptation because it helps people adjust to flood-prone conditions.
3	D	Mangroves reduce flooding risks (adaptation) and store carbon dioxide (mitigation), so it counts as both.
4	B	Planting drought-resistant crops is the best adaptation strategy for surviving frequent droughts.
5	D	Healthy ecosystems provide food, water, protection, and services needed for human survival.
6	C	Cutting trees causes soil erosion, which leads to floods downstream.
7	D	Solar streetlights and waste segregation directly reduce emissions, making them mitigation strategies.
8	D	Adaptation is urgent because communities are already facing climate impacts today.
9	D	Adaptation is equally important because it allows people to survive current climate threats.
10	D	Planting trees in watersheds is a community adaptation practice that reduces flooding while supporting biodiversity.
11	D	Without mangroves, coastal areas lose fish habitats and become more prone to flooding.
12	C	Composting and organic fertilizers reduce emissions (mitigation) and improve soil resilience (adaptation).
13	D	Urban heat islands trap heat and worsen air quality, which harms both biodiversity and people.
14	B	Seawalls are adaptation, and solar power is mitigation — combining both addresses current and future risks.
15	D	Reforestation plus disaster education is sustainable because it protects ecosystems and prepares communities.